

ROBERT SIMON FONG

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RESEARCH INTERESTS

- Differential Geometry, Information Geometry: theory and applications in Manifold Optimization
- Symplectic Geometry and Representational Learning
- Machine Learning and Dynamical Systems: Reservoir Computing
- Black-box Optimization: Bayesian Optimization, Zeroth-order Optimization

EDUCATION

University of Birmingham

Sept. 2015 - July 2020

Ph.D., Computer Science

Thesis: Stochastic Optimization on Riemannian Manifolds

Supervisors: Prof. Peter Tiño, Prof. Joshua Knowles

University of Waterloo

Sept. 2013 - Aug. 2014

Master of Mathematics, Computational Mathematics

Thesis: Optimization with Function Values Only

Supervisor: Prof. Thomas Coleman

University of Waterloo

Sept. 2007 - Apr. 2012

Bachelor of Mathematics, Pure Mathematics (Honours)

ACADEMIC POSITIONS

Post-Doctoral Research Fellow (特任研究員), University of Tokyo

May. 2025 - present

- **Research:** Symplectic Reservoir Computing
- Funded the International Research Center for Neurointelligence (WPI-IRCN) at The University of Tokyo Institutes for Advanced Study

Research Fellow, University of Birmingham

Feb. 2021 - present

- **Research:** Reservoir Computing: Universality and applications on time series forecasting.
- **Teaching:** Mathematical Foundations of Artificial Intelligence and Machine Learning (MSc) [06 32250], Algorithms for Data Science (MSc) [06 32258]
- Funded by Alan Turing Institute; Prof. Peter Tiño's Alan Turing Institute Fellowship
Machine Learning in the Space of State-Space Dynamic Models

Ph.D. Candidate, University of Birmingham

Sept. 2015 - Jul. 2020

- **Research** Manifold Optimization using Differential Geometry, Information Geometry and Simplicial Geometry.
- **Teaching:** MSc/ICY Software Workshop (Java) (1,2), Mathematical Foundations of Computer Science

Master Student, University of Waterloo

Sept. 2013 - Aug. 2014

- **Research:** Derivative-free optimization: Simulated Annealing and Derivative-Free Zeroth-Order surrogate directional search methods. (with Prof. Thomas Coleman)

- **Research:** Computational Algebra: Solving multivariate polynomials using diagonal subgroup of linear group action. (with Prof. George Labahn)
- **Teaching:** Math 106 Linear Algebra for Arts (Co-organized).

Undergraduate Research Assistant, University of Waterloo

Jan. 2012 - Apr. 2012

- Integer Programming and applications on Quay Crane Scheduling problem (special case of Traveling Salesman problem)

Undergraduate Research Assistant, Penn State University

Jan. 2009 - Mar. 2009

- Optimization of Gravitational Wave Detectors in Laser Interferometer Gravitational-Wave Observatory (LIGO).

INDUSTRY EXPERIENCE

Senior Researcher, Theory Lab, Huawei

Aug. 2022 - Mar. 2024

- Quantum-Inspired Optimization – theory of Hamiltonian-based solver and its applications:
 - Boolean Satisfiability Problem, Electronic Design Automation (EDA), Hardware Verification, Graph partitioning, and MIMO Decoding.
- Reservoir Computing-based Time Series Forecasting
- Hardware Placement and Routing modelling in Integrated Circuits

Researcher, Noah's Ark Lab, Huawei

Aug. 2020 - Aug. 2021

- Theory of Deep Neural Nets: convexity, modelling, and optimization
- Application and implementation of Deep Neural Nets and Bayesian Optimization:
 - 5G+ High Frequency Surface antenna design and Integrated Circuit Partition.

Research Analyst, Cayuga Research Associates

Jan. 2013 - Aug. 2015

- Optimization and Simulation of Container Port control system.

Consultant, Hong Kong International Terminals Limited

Apr. 2012 - Jan. 2013

- Integer Programming Modelling and Abstraction of ISO container placement

BOOKS AND MONOGRAPHS

- **Robert Simon Fong** and Peter Tiño. *Population-Based Optimization on Riemannian Manifolds*, volume 1046 of *Studies in Computational Intelligence (SCI)*. Springer, 2022. ISBN: 978-3-031-04292-8 (eBook ISBN: 978-3-031-04293-5)

SELECTED JOURNAL ARTICLES

- **Robert Simon Fong**, Gouhei Tanaka, and Kazuyuki Aihara. Symplectic reservoir representation of legendre dynamics, 2025
- **Robert Simon Fong**, Boyu Li, and Peter Tiño. Linear simple cycle reservoirs at the edge of stability perform fourier decomposition of the input driving signals. *Chaos*, 35(4):043109, 2025
- **Robert Simon Fong**, Yanming Song, and Alexander Yosifov. Efficient digital quadratic unconstrained binary optimization solvers for sat problems. *New Journal of Physics*, 27(1):013027, Jan 2025
- Boyu Li, **Robert Simon Fong**, and Peter Tiño. Simple Cycle Reservoirs are Universal. *Journal of Machine Learning Research*, 25(158):1–28, 2024

- Robert Simon Fong. *Stochastic optimization on Riemannian manifolds*. PhD thesis, University of Birmingham, 2020

SELECTED CONFERENCE PAPERS(REFERRED)

- Ziqiang Li, **Robert Simon Fong**, Kantaro Fujiwara, Kazuyuki Aihara, and Gouhei Tanaka. Structuring Multiple Simple Cycle Reservoirs with Particle Swarm Optimization. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–9, 2025
- **Robert Simon Fong**, Boyu Li, and Peter Tino. Universality of real minimal complexity reservoir. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(16):16622–16629, 2025
- Peter Tiño, **Robert Simon Fong**, and Roberto Fabio Leonarduzzi. Predictive Modeling in the Reservoir Kernel Motif Space. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2024
- **Robert Simon Fong** and Peter Tiño. Extended stochastic derivative-free optimization on riemannian manifolds. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, GECCO '19, pages 257–258, New York, NY, USA, 2019. ACM
- **Simon Fong** and Peter Tiño. Induced dualistic geometry of finitely parametrized probability densities on manifolds, 2018
- David Tsang, Andrew Lundgren, Ruxandra Bondarescu, **Simon Fong**, and Mihai Bondarescu. Optimizing finite mirrors for advanced gravitational wave detectors. In *APS April Meeting Abstracts*, 2009

AWARDS AND SCHOLARSHIPS

IRCIN Retreat Brainstorming Award FY2025 , IRCIN, The University of Tokyo	2025
Honorary Research Fellowship , University of Birmingham	2021 - 2026
President's Scholarship , University of Waterloo	2008
AP Scholar with Distinction , The College Board	2007
Second Honour , Third Pan Pearl Delta plus Chinese Elite Schools Physics Olympiad	2007
Third Honour , Hong Kong Physics Olympiad	2007
Third Honour , Hong Kong Physics Olympiad	2006

INVITED TALKS

AI Incubator Salon, IRCIN, The University of Tokyo , Tokyo, Japan	9. Jul. 2025
<i>Universality of Simple Cycle Reservoirs</i>	
Invited Talk	
Drexel University , Philadelphia, PA, USA	28. Feb. 2025
<i>Universality of Simple Cycle Reservoirs and Dilation Theory</i>	
Invited Talk with Prof. Boyu Li	
Southern University of Science and Technology , Shenzhen, China	23. Aug. 2024
<i>Universality of Simple Cycle Reservoirs</i>	
Invited Talk	
Huawei Noah's Ark Lab , Hong Kong, China	15. Mar. 2020
<i>Information Geometry of Variational Autoencoder</i>	
Invited Talk	

JP NATIONAL CONFERENCE TALKS

JST 若手数学者交流会 (第 7 回) , Tokyo, Japan	21. Mar. 2026
<i>Symplectic Reservoir Representation of Legendre Dynamics</i>	
Poster + Talk	
第 3 回バイオ数理連携会議 , Tokyo, Japan	04. Dec. 2025
<i>Simple Cycle Reservoirs: Universality and Applications</i>	
Poster	
ICNSM2025 , Tokyo, Japan	19. Nov. 2025
<i>Harmonics at the Edge of Stability</i>	
Poster	
IRCIN Retreat 2025 , Odawara, Japan	23. Oct. 2025
<i>Simple Cycle Reservoirs: Universality in real numbers</i>	
Poster + Talk	
IRCIN Retreat 2025 , Odawara, Japan	23. Oct. 2025
<i>A Particle Swarm Optimization Approach to Structuring Multiple Simple Cycle Reservoirs</i>	
Poster + Talk, with Z. Li, K. Fujiwara, K. Aihara, and G. Tanaka	
IRCIN WPI site visit 2025 , Tokyo, Japan	16. Jul. 2025
<i>Harmonics at the Edge of Stability</i>	
Poster + Talk	
IRCIN WPI site visit 2025 , Tokyo, Japan	16. Jul. 2025
<i>A Particle Swarm Optimization Approach to Structuring Multiple Simple Cycle Reservoirs</i>	
Poster + Talk, with Z. Li, K. Fujiwara, K. Aihara, and G. Tanaka	

RESEARCH GRANTS

IRCIN Retreat Brainstorming Awards FY2025, JPY 800,000	2025-2026
Principal Investigator , 3 partners	
<i>Multi-Reservoir Approach to Modular Invariance and Conservation in Biological Networks</i>	
AI Incubator Startup Grant, JPY 1,000,000	2025-2027
Principal Investigator	
<i>Hamiltonian Representation in Reservoir Computing</i>	

REVIEW SERVICES

- SIAM Journal on Optimization
- ACM/SIGEVO Conference on Foundations of Genetic Algorithms (FOGA)